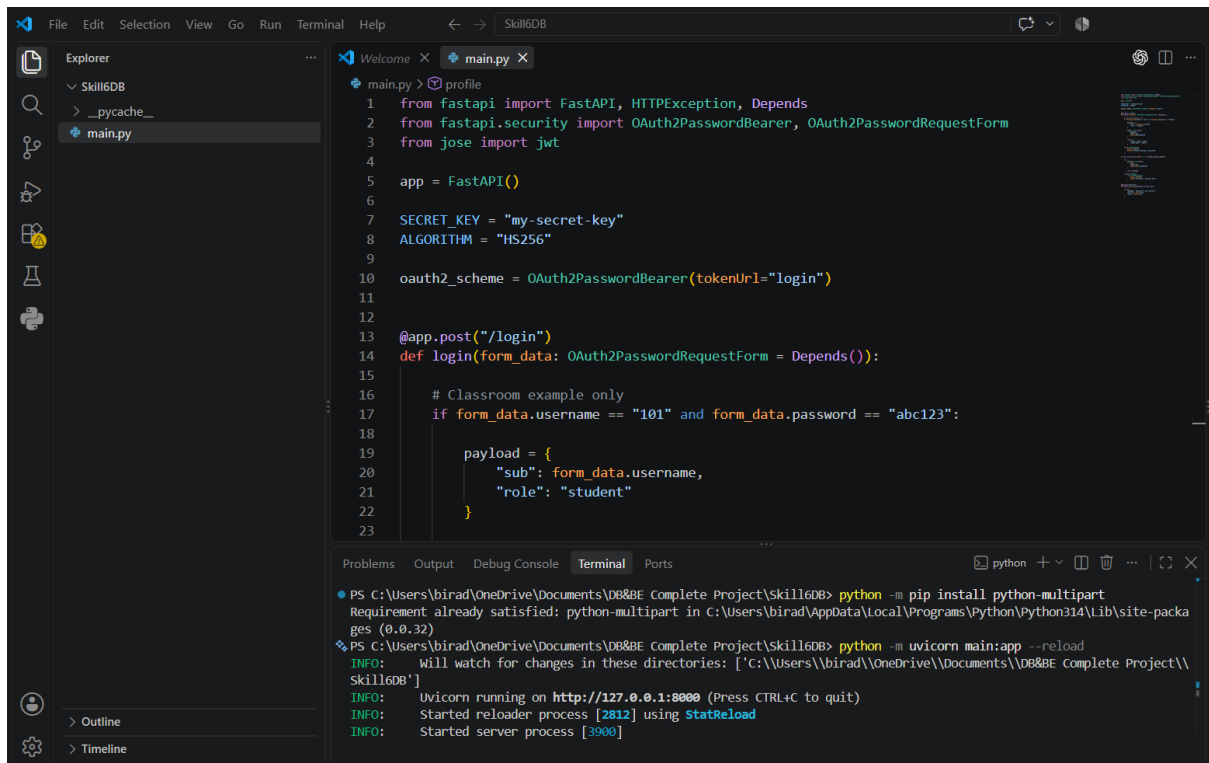


2520090008_Sathish kumar

Skill – 6



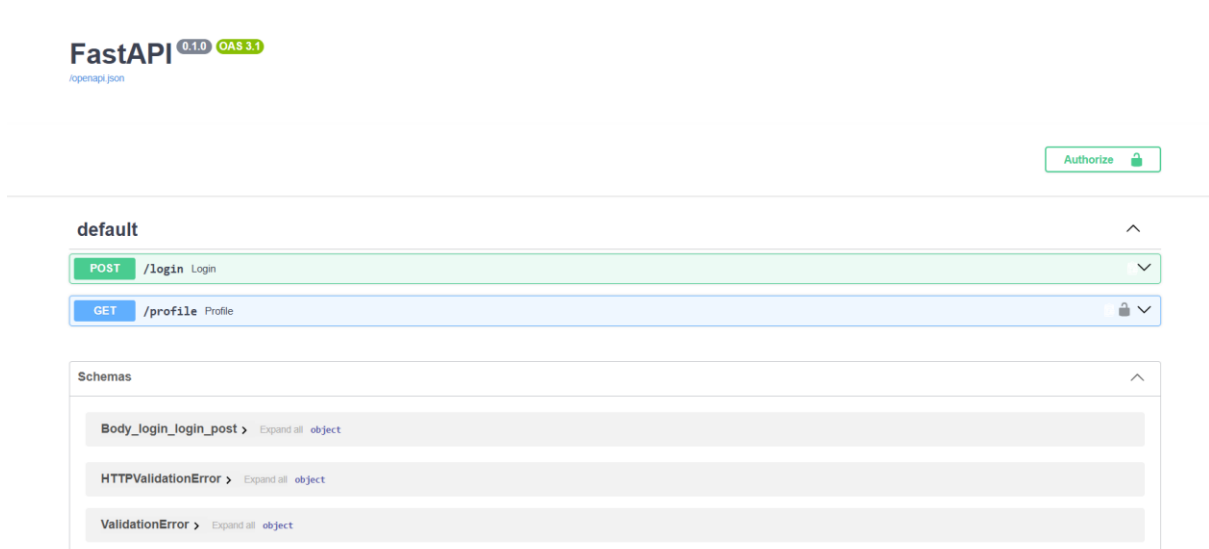
The screenshot shows a VS Code editor with a project named 'Skill6DB'. The Explorer panel on the left shows the file structure with 'main.py' selected. The main editor displays the code for 'main.py', which is a FastAPI application. The code includes imports for FastAPI, HTTPException, Depends, OAuth2PasswordBearer, OAuth2PasswordRequestForm, and jwt. It defines a FastAPI app, sets a secret key and algorithm, and creates an OAuth2PasswordBearer instance. A POST endpoint '/login' is defined, which checks the username and password and returns a JWT token if they match. A GET endpoint '/profile' is also defined. The terminal at the bottom shows the command to run the application: 'python -m uvicorn main:app --reload'. The output shows that the application is running on http://127.0.0.1:8000.

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15
16     # Classroom example only
17     if form_data.username == "101" and form_data.password == "abc123":
18
19         payload = {
20             "sub": form_data.username,
21             "role": "student"
22         }
23
```

Terminal Output:

```
PS C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB> python -m pip install python-multipart
Requirement already satisfied: python-multipart in C:\Users\birad\AppData\Local\Programs\Python\Python314\Lib\site-packages (0.0.32)
PS C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\birad\OneDrive\Documents\DB&BE Complete Project\Skill6DB']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [2812] using StatReload
INFO: Started server process [3900]
```

Swagger:



POST/Login:

default

POST /login Login

Parameters

Try it out

No parameters

Request body required

application/x-www-form-urlencoded

grant_type
string | (string | null)
pattern: ^password\$

username * required

string

password * required

string(\$password)

scope
string

client_id
string | (string | null)

client_secret
string | (string | null)(\$password)

Authorize in Swagger:

astAPI 0.1.0 OAS 3.1

default

POST /login Login

Parameters

Cancel Reset

No parameters

Request body required

application/x-www-form-urlencoded

grant_type
string | (string | null)
pattern: ^password\$

username * required

string101

password * required

string(\$password)*****

scope
string

☒ Send empty value

Available authorizations

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes. API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Token URL: login

Flow: password

username:

password:

Client credentials location:

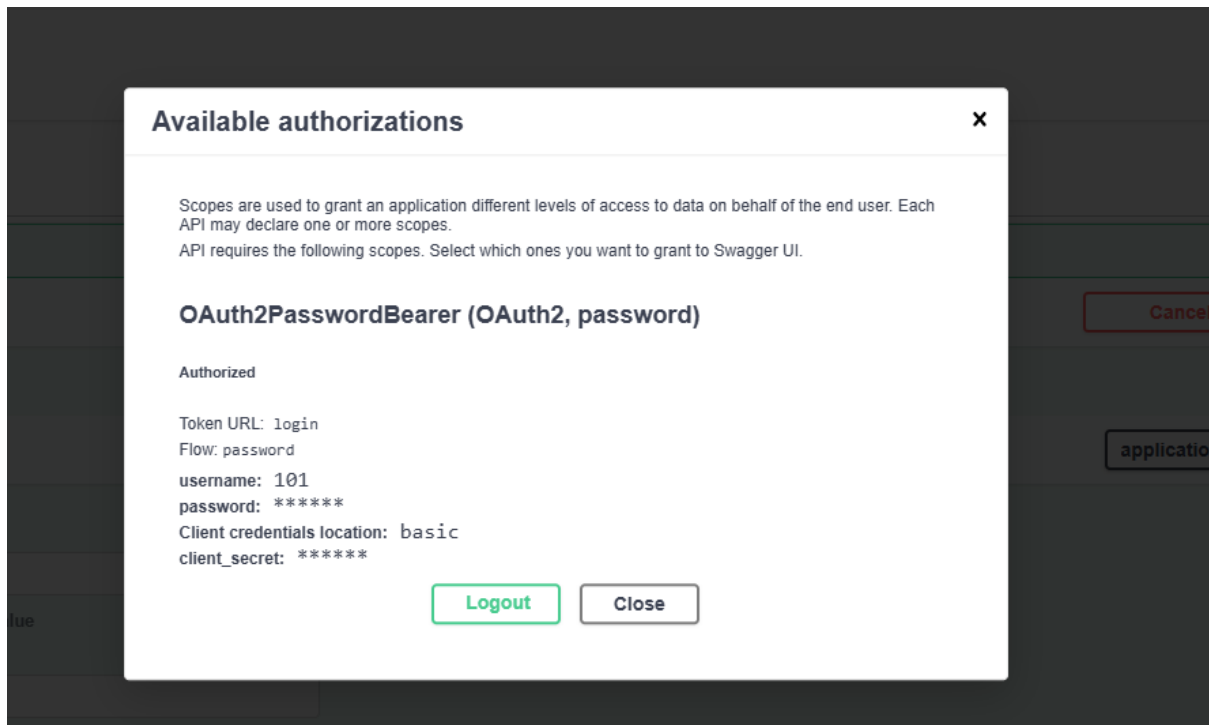
Authorization header

client_id:

client_secret:

Authorize

Close



GET/Profile:

GET

/profile Profile

🔒 ⤴

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema "string"	No links

GET

/profile

Profile



Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVjQ9LWVudC39.NHqwDLPBiq5BFQ1KmdoWvL0jblUKWtH4aRQ6MK1zI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code	Details
200	Response body <pre>{ "message": "Welcome to your profile", "user_id": "101", "role": "student" }</pre>  Download Response headers <pre>content-length: 70 content-type: application/json date: Fri, 18 Sep 2026 04:07:00 GMT server: uvicorn</pre>

Responses

Code	Description	Links
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